

M.AI.ESTRO-Twin Vision Pipeline Placement Report

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Unit ARI5907 Placement in AI

Placement Title M.AI.ESTRO-Twin: Automated Digital Twin Synthesis for Industrial Skill Transfer

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Abstract

This report presents the final outcome of the ARI5907 placement, which contributed a reusable vision perception layer to the M.AI.ESTRO-Twin project. The work addressed how to generate structured perception signals from egocentric manipulation demonstrations without manual prompt entry. Qwen3-VL-2B-Instruct paired with SigLIP was selected as the production semantic configuration, offering the best quality-to-latency ratio across six evaluated models at 1.432 seconds per window. Semantic object lists were then used as automated prompts for Grounding DINO and SAM 2. A frozen 512-window state-change evaluation across EgoDex training parts 1–3 produced 97.3% window-level detection and mask coverage at the 0.25 detector threshold, measured as proxy coverage rather than annotated precision or recall. The resulting manifests contain captions, tags, object prompts, embeddings, bounding boxes, and masks suitable for downstream world model and policy conditioning. The 6 DoF stage was deferred because the semantic and 2D grounding stages had reached a useful prototype state, while geometry-aware pose recovery would have required substantially heavier CAD and evaluation assumptions.

1 Introduction

The placement was carried out alongside Research Support Officer duties linked to Project M.AI.ESTRO, an active Malta based research programme developing automated digital twin technology for industrial skill transfer. The declared placement title remained *M.AI.ESTRO-Twin: Automated Digital Twin Synthesis for Industrial Skill Transfer*, and the motivating problem was how to generate structured perception signals automatically from expert demonstrations without the need for manual prompt entry or annotation.

Two related problems motivated the study. The first was the broader representation problem faced by video and world model pipelines. These systems compress raw visual streams into latent state representations, but raw video includes task relevant dynamics alongside nuisance variation such as camera motion, lighting change, background motion, and object movements that are not conditioned on the target skill. This creates a familiar shortcut-learning risk in visual modelling: a system can rely on visually available but task-irrelevant correlations when those correlations are easier to encode than the intended causal structure [1]. In a dash camera setting, for example, a poorly conditioned representation may spend capacity modelling sky, tree, or camera-motion regularities rather than the traffic behaviour needed

for driving. In the placement context, object lists, action descriptors, boxes, and masks were therefore treated as enrichment signals that help focus downstream learning on the manipulated objects and task relevant state changes.

The second problem was more implementation specific. Open-set perception tools can localise arbitrary text-conditioned objects, but they normally require a prompt or object list before localisation begins. In a manually operated workflow, that prompt would be supplied by a human annotator or user. Figure 1 illustrates this prompt-conditioned stage. The placement addressed this bottleneck by using the vision language model semantic stage to generate the object list automatically, so that the same sampled EgoDex window could drive semantic description, object prompting, grounding, and mask generation as a unified automated pipeline.

The one page progress report submitted at the midpoint of the placement established that a working semantic alignment pipeline had been implemented and that an initial vision language model comparison had been completed. This final report continues from that checkpoint by documenting how the semantic stage was refined, how downstream grounding and mask stages were added, and why the placement delivered a 2D perception and conditioning layer rather than the full four stage pipeline originally proposed.

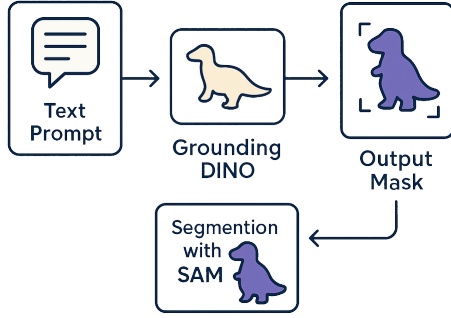


Figure 1: Prompt-conditioned grounding problem addressed by the placement. Open-set detectors can localise text-specified objects, but the text prompt normally has to be supplied before detection can begin.

The aim of this placement was to develop a reusable AI vision pipeline that transforms raw egocentric manipulation data into machine readable representations of task relevant state, suitable for conditioning downstream learning modules within M.AI.ESTRO. To achieve this aim, five objectives were pursued:

1. Establish a semantic alignment pipeline capable of generating structured object and action descriptors from EgoDex manipulation windows.
2. Evaluate and select a vision language model and embedding backend appropriate for automated semantic quality assessment.
3. Extend the semantic stage into a downstream grounding pipeline using open-set object detection and segmentation.
4. Produce aligned structured manifests containing captions, tags, object prompts, bounding boxes, and masks suitable for world model conditioning.
5. Document the design decisions and model selection rationale in a form that supports future M.AI.ESTRO pipeline development.

The central research question guiding the implementation was: *Can vision language model object lists serve as reliable automated prompt sources for downstream open-set object grounding in egocentric manipulation data?* The original proposed stages were semantic understanding, bounding box grounding, binary mask generation, and 6 DoF pose estimation. As the implementation matured, the scope was deliberately narrowed to the first three stages because they formed the perception enrichment layer most directly needed by the downstream world model work. The report is structured as follows:

- Section 2 describes the methodology.
- Section 3 presents the results.
- Section 4 reports placement effort.
- Section 5 draws conclusions.
- Section 6 identifies future work.

2 Methodology

All data used in this placement was drawn from EgoDex, a publicly released dataset; no human participants were recruited, no personal data were collected, and no ethical approval beyond standard academic data use was required. EgoDex is a large scale egocentric manipulation dataset comprising approximately 1.1 million demonstration clips across 194 object categories designed for dexterous interaction learning [2]. The placement did not process the full dataset. It used a local subset drawn from EgoDex training parts 1, 2, and 3.

The first implemented windowing version was deliberately simple. Each sample contained 16 observation frames followed by 16 prediction frames, and candidate windows were produced using a fixed stride of 128 frames. At 30 fps, this placed consecutive candidate starts approximately 4.3 seconds apart. This was sufficient for the initial semantic benchmark, where a shared 64-window slice was drawn from the fixed index and held constant across all candidate generators and embedding backends. Its purpose was model ranking rather than dataset-level performance estimation.

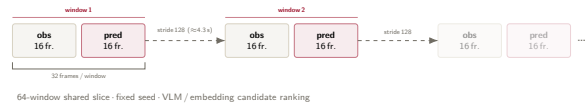


Figure 2: Fixed-stride windowing used for the semantic bakeoff. Each window spans 16 observation frames followed by 16 prediction frames; consecutive starts are 128 frames apart (≈ 4.3 s at 30 fps). The 64-window shared slice was held constant across all candidate generators and embedding backends. Author-generated schematic.

The live evaluation method used for the final grounding and mask results is more selective. Instead of taking fixed-stride windows alone, the sampler discovers paired HDF5 and video episodes across train parts 1–3, computes smoothed hand-state change scores from `transforms/leftHand`, `transforms/rightHand`, `confidences/leftHand`, and `confidences/rightHand`, selects candidate windows near high-change regions, and keeps relative-position fallback candidates for slower episodes. It then samples episode-first and task-stratified with seed 42, yielding one frozen window

per episode. The final evaluation subset contains 512 distinct episodes across 63 tasks, with 151 windows from train part 1, 187 from train part 2, and 174 from train part 3. The selected window IDs, JSONL inventory, and subset pickle were saved so the run can be repeated exactly. Figure 2 shows the first fixed-stride version, while Figure 3 shows the current state-change-targeted evaluation method.

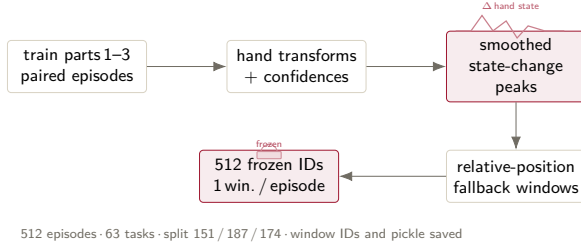


Figure 3: State-change-targeted evaluation sampling. Episodes from train parts 1–3 are scored by smoothed hand-state change signals; relative-position fallback candidates retain slower episodes; episode-first, task-stratified sampling (seed 42) yields one frozen window per episode across 512 episodes and 63 tasks. Author-generated schematic.

2.1 Semantic Alignment

The first semantic candidate used during development was Qwen3-VL-8B-Instruct, selected because the early project goal required a model capable of generating scene level descriptions and structured semantic signals from a small set of representative frames [3]. The initial implementation was tightly coupled to that model, with prompt templates, output parsing, and quality checks written specifically for its response format. Once the semantic stage was shown to function reliably end-to-end, the code was refactored into a model agnostic benchmark contract with a shared prompt interface, standardised output schema, and interchangeable embedding backend. This refactoring was essential because the objective was not simply to make the pipeline run once, but to produce a reusable and comparatively inspectable semantic stage.

In this context, embeddings are the vector representations used to place text and visual content into a common comparison space [4]. The semantic generator produced captions and structured tags, while the embedding backend was used to assess how well those generated descriptions still corresponded to the sampled window frames. The embedding model was therefore not merely an implementation detail, but part of the semantic quality check loop itself. SigLIP became the main operational backend because its pairwise sigmoid training objective produces per-pair similarity scores that do not depend on batch composition, mak-

ing it well suited to the role of an independent semantic quality check on individual window caption pairs [5]. The project later also explored OpenCLIP as an alternative similarity model in order to test whether the semantic ranking remained stable under a different comparison space [6].

The candidate generator set then widened. The Qwen family was compared with LLaVA-v1.6 [7], SmolVLM2-2.2B-Instruct [8], and Idefics3-8B [9]. During this benchmarking stage, the release of the Qwen 3.5 family provided an opportunity to extend the comparison further and examine whether the new variants changed the quality versus latency trade off enough to justify a change in the operational pipeline [10, 11]. These later checks became part of the placement process rather than a separate line of work.

The output schema of the semantic stage was refined iteratively as the pipeline moved from human readable inspection artefacts to machine parseable structured manifests. Initially, the stage emphasised raw artefacts so that the output could be inspected directly and used to judge whether the model was describing the task window sensibly. As the placement moved towards a multi stage pipeline, human readable artefacts alone were no longer sufficient: the grounding stage required a consistent, machine parseable object list rather than free text captions to function without manual intervention. The outputs therefore had to be written in a structured schema so that the next stage could consume them reliably. For this reason, the semantic stage was revised to write an aligned manifest, captions, tags, embeddings, and especially the object list information that acts as the prompt source for downstream grounding. The remaining manifest fields are also useful later as richer conditioning signals, particularly when contact relations or state changes are detected in the window.

2.2 Bounding Box Grounding

Once the semantic stage had stabilised, its outputs were used to drive the next stage of the pipeline. Rather than entering text prompts manually, prompt candidates were derived from the semantic results themselves, with particular attention to the contact object and the deduplicated object list. This design choice preserved the pipeline logic established in the semantic stage: the model first interpreted the task window and then supplied the vocabulary required for localisation.

Those prompts, exemplified in Figure 4, were passed into Grounding DINO, which produced text conditioned bounding boxes [12]. This step was important because the placement was not limited to scene description and had to move from seman-

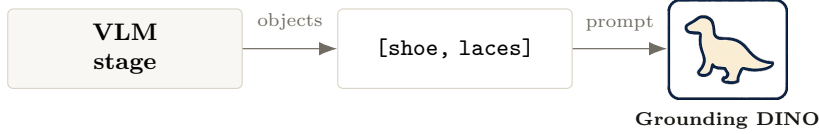


Figure 4: Simplified prompt handoff used by the implemented pipeline. The semantic stage emits a compact object list, and that list becomes the text prompt for Grounding DINO.

tic interpretation towards explicit image space localisation. The resulting boxes therefore became the first direct spatial outputs produced by the implemented pipeline. Grounding DINO was selected on the basis of published benchmark performance rather than through an internal benchmark as extensive as the semantic VLM bakeoff. At the time of selection it represented the state of the art on zero-shot open-set detection, with published results reporting 52.5 AP on COCO without any in distribution training images and a mean 26.1 AP on the ODinW zero-shot benchmark, outperforming prior alternatives including GLIP and OWL-ViT [12]. In use, the stage showed the expected weaknesses of zero-shot grounding: small or visually similar objects could be confused, and category labels were not always assigned consistently. Nevertheless, the detector generally placed boxes in the correct image regions, which was the main requirement at this stage because those regions were then passed forward to mask generation. The archived run used the Hugging Face Grounding DINO post-processing default threshold of 0.25 for both box and text filtering, while the stricter 0.35 box threshold reported below follows the official Grounding DINO example configuration as a post-hoc sensitivity check [13, 14].

2.3 Object Mask Generation

The final implemented stage of the placement used SAM 2 after the bounding box step so that the detected objects also yielded binary masks [15]. This gave the pipeline denser spatial outputs than boxes alone and made the results more useful for later conditioning and inspection. The purpose of this stage was not to recover geometry or stable 3D identity. Its role was to provide a practical 2D segmentation signal aligned to the same windows already used by the semantic and grounding stages.

In practical terms, integrating SAM 2 required implementing three concrete components: a box expansion step that enlarged detector outputs by a configurable pixel margin before passing them to the segmenter, a frame level batching loop that applied SAM 2 to each observation frame in the window, and a run length encoding serialiser that stored the resulting masks compactly together

with their spatial shape metadata. Just as importantly, the mask rows remained aligned with the original semantic window order, which preserved a stable correspondence between captions, tags, detections, and masks for downstream reuse. SAM 2, which extends the original Segment Anything Model [16] to video and streaming contexts, was chosen largely from the literature and from its strong practical reputation as a general purpose segmentation model, rather than from a dedicated in house benchmark across many maskers.

The known limitation of this stage, category confusion between visually similar objects, is quantified in Section 3.2 and listed in Section 3.3.

3 Results

3.1 Semantic Benchmark Results

The semantic benchmark expanded the model comparison first introduced in the one page progress report. Six candidate generators were evaluated: Qwen3-VL-2B, Qwen3-VL-8B, LLaVA-v1.6-7B, Idefics3-8B, Qwen3.5-4B, and SmolVLM2-2.2B. The evaluation sample consisted of 64 windows drawn deterministically from the EgoDex index using a fixed random seed, stratified to include a range of object categories and task phases. The benchmark slice was held fixed across all runs, with only the embedding backend varied, so that the contribution of the generator could be separated cleanly from the contribution of the similarity model. The sample size was appropriate for ranking candidate configurations rather than estimating absolute dataset-level performance; the OpenCLIP cross-family check was included to test whether the relative ordering remained stable under a second embedding space.

Table 1 summarises the main outcome across all evaluated generators. Both SigLIP and OpenCLIP scores are reported for each model.

Table 1: Semantic benchmark results over the shared 64-window sample. SigLIP score and OpenCLIP score are cosine similarity proxies computed with each embedding family independently; they are not directly comparable across families. Lower seconds per window indicates lower runtime cost.

Model	SigLIP score	OpenCLIP score	Sec./win
Qwen3-VL-2B	0.365	0.409	1.432
Qwen3-VL-8B	0.431	0.473	2.435
Qwen3.5-4B	0.434	0.478	5.763
LLaVA-v1.6-7B	0.422	0.463	3.317
Idefics3-8B	0.405	0.450	4.186
SmolVLM2-2.2B	0.319	0.361	7.046

The comparison led to two practical results. First, SigLIP remained the benchmark-backed operational choice. The one page checkpoint and the final implementation both used SigLIP as the main embedding backend because its pairwise scoring behaviour is better suited to judging individual image-text pairs. OpenCLIP was retained only as a comparative check. Its scores were consistently higher than the matching SigLIP scores, but that does not mean the generated captions were better: OpenCLIP and SigLIP use different contrastive objectives and produce values on different scales [5, 4]. The OpenCLIP column therefore shows that the model ranking was broadly stable under a second embedding family, not that the pipeline should switch away from SigLIP.

Second, the Qwen family remained the strongest practical choice once quality and latency were considered together. Qwen3-VL-2B returned the lowest absolute SigLIP score in the benchmark (0.365), but a score at this level remains a strong semantic alignment result for egocentric manipulation text-image pairs, and the model achieves this at 1.432 s/win compared to 2.435 s/win for the next model and over 5 s/win for the top-scoring variants. For a pipeline intended to scale across large data extractions, the $1.7\times$ to $5\times$ inference advantage outweighs the marginal quality gap, making Qwen3-VL-2B-Instruct together with SigLIP the most practical production choice. Figure 5 shows the quality versus runtime trade off that supported the final operational choice.

3.2 Grounding and Mask Results

The grounding and mask stages produced the first explicit spatial outputs from the placement pipeline. Using the semantic object list as the prompt source, Grounding DINO generated text conditioned bounding boxes, and SAM 2 then produced the corresponding binary masks. These results confirmed that the semantic stage was not only interpretable in isolation, but also useful as a prompt generator for downstream perception modules. Figure 6 shows an example output, while Figure 7 shows a second qualitative case with a

different object configuration.

The strongest positive result here was that the semantic-to-grounding pipeline could be automated without freezing on manual prompt entry. To quantify this behaviour without creating new manual annotations, the selected Qwen3-VL-2B/SigLIP semantic pipeline was rerun over the frozen 512-window evaluation subset described in Section 2 and then passed through Grounding DINO and SAM 2.

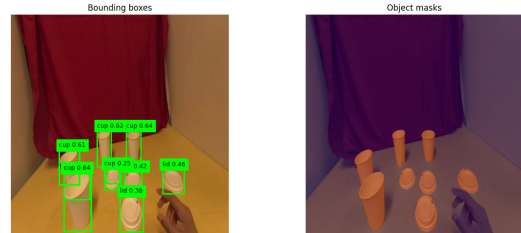


Figure 6: Example placement output from the grounding and mask stages. Semantic object prompts become bounding boxes and binary masks for the sampled window frames.

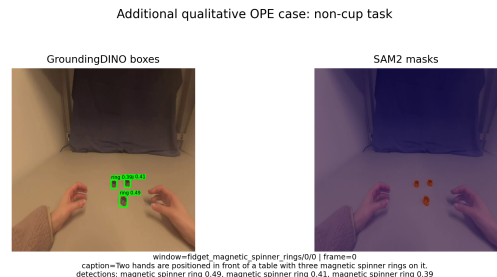


Figure 7: Additional qualitative example from a ring handling episode, showing a second object configuration beyond the cup case in Figure 6.

At the 0.25 threshold, 498 of 512 windows produced at least one bounding box and corresponding mask, with detections present in 980 of 1024 sampled frames. The stricter 0.35 filter retained 469 windows and 888 frames, confirming the result was not driven by borderline detections. Grounding DINO produced 2,208 boxes at 0.25, of which 1,405 remained above 0.35, and prompt-label coverage held at 96.2% across all 63 task groups. The main failure pattern was category confusion between visually similar objects: 51 windows at 0.25 and 34 at 0.35 contained overlapping boxes assigned to different prompt labels at IoU greater than 0.5.

Table 2: Grounding and mask proxy results over the frozen 512-window state-change evaluation subset. The 0.25 row reports the detector threshold used during inference, while the 0.35 row applies a stricter post-hoc score filter. These metrics measure whether the automated prompts produced usable spatial outputs; they are not precision, recall, or IoU against ground truth annotations.

Threshold ≥ 0.25	
Window detection rate	97.3%
Frame detection rate	95.7%
Mean top-1 box score	0.547 ± 0.144
Mask window rate	97.3%
Mean SAM 2 mask score	0.905 ± 0.105
Threshold ≥ 0.35	
Window detection rate	91.6%
Frame detection rate	86.7%
Mean top-1 box score	0.562 ± 0.135
Mask window rate	91.6%
Mean SAM 2 mask score	0.912 ± 0.104

No formal precision or recall metric was computed against ground truth annotations, which remains a limitation of the current evaluation. The sample is nonetheless frozen, cross-part, state-change targeted, and fully reproducible from saved window identifiers. The proxy results therefore support the central operational claim directly: semantic object lists are a practically viable automated prompt source, giving the placement a coherent multi-stage workflow rather than a set of disconnected experiments.

3.3 Limitations

Four principal limitations bound the scope of these results. First, the semantic benchmark was a bounded comparison over a 64-window shared sample; it is not a universal ranking and may not generalise to object categories or task types not represented in the sample. Second, the grounding and mask stages were evaluated using frozen

proxy metrics and qualitative inspection rather than against a formal annotated validation set with ground truth boxes or masks; no precision, recall, or IoU figures are reported. Third, the placement pipeline was developed and tested exclusively on EgoDex; cross dataset generalisation to other egocentric manipulation datasets was not evaluated. Fourth, although the pipeline produced enrichment data intended to reduce nuisance variation in downstream latent state learning, the placement did not yet train or evaluate a world model using those enriched signals. The report can therefore claim that the enrichment layer was produced and aligned, but not that it has already solved the noisy latent representation problem introduced in Section 1.

3.4 Interpretation

This bounded reading also explains the 6 DoF decision. By the time the semantic, bounding box, and mask stages had matured into a coherent prototype, the next step towards canonical 6 DoF pose recovery would have required a substantially different level of geometric machinery, stronger CAD dependence, and a more specialised evaluation setup [17, 18]. The placement therefore reached its best value by consolidating the 2D perception and conditioning pipeline rather than pushing prematurely into a heavier geometric stage.

4 Placement Effort and Supervision

The placement unit corresponds to approximately 100 to 140 hours of work. The hours worked over the placement period can be summarised as shown in Table 3, which presents a reconstructed allocation of 130 hours across the completed stages of the implementation. The allocation is estimated rather than logged: a conservative basis of 5 hours per day across 26 placement equivalent days was

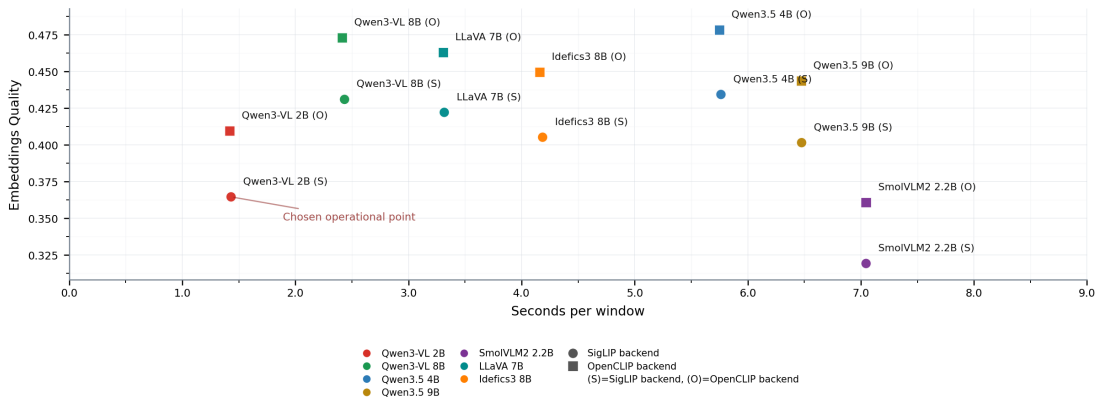


Figure 5: Semantic quality versus runtime trade off used to support the final production choice.

used, cross referenced against calendar records and implementation commit history. These activities were not always sequential, since model runtime, reruns, and waiting periods were interleaved with implementation, inspection, and refinement across multiple stages of the pipeline.

Table 3: Retrospective placement effort allocation reconstructed from the completed implementation stages.

Placement activity	Days	Hours
Environment preparation, Docker checks, and <code>kr2280</code> setup	1	5
Semantic pipeline implementation and early validation	6	30
Object grounding implementation and prompt construction	5	25
Mask generation integration and refinement	6	30
Model rebenchmarking and later candidate checks	4	20
Notebook walkthroughs, result consolidation, and visual inspection	2	10
Report preparation and final packaging	2	10
Total	26	130

This placement was also shaped by continuous supervision. The implementation state, benchmark outcomes, and main design decisions were communicated regularly to Prof. Alexiei Dingli throughout the placement period. This supervisory loop proved particularly important when new candidate models became available and when benchmark deltas required careful interpretation before any change to the operational pipeline was warranted.

5 Conclusion

The placement successfully moved from raw EgoDex windows to structured semantic outputs, image space localisation, and binary masks, producing a modular, benchmark driven pipeline that can be extended or replaced at the component level. Returning to the central research question, the evidence confirms that vision language model object lists can serve as reliable automated prompt sources for downstream open-set object grounding in egocentric manipulation data: semantic object lists generated by Qwen3-VL-2B-Instruct triggered consistent detections across the frozen 512-window evaluation sample without any manual prompt entry.

The value of this approach is further reinforced by its alignment with current research practice. DexWM [19] encodes scene states using a frozen DINOv2 encoder whose semantically rich patch-level features serve the same enrichment role as

the structured outputs produced here. PointWorld [20] independently enriches raw point clouds with frozen DINOv3 ViT-L visual features before world model conditioning. The enrichment strategy developed in this placement is therefore not an incidental design choice but a practice convergently validated at the frontier of the field.

6 Future Work

Three immediate directions follow from this placement. First, the semantic and open-set perception stages should be framed explicitly as a world model enrichment layer, exposing captions, object lists, boxes, and masks as a structured interface for downstream predictors. Second, that layer can support an enriched world model variant for M.AI.ESTRO, testing whether semantic prompts, spatial detections, and masks improve predictive representations for industrial skill transfer in the spirit of DexWM and PointWorld. Third, the methodology and reusable code are applicable to the Malta–Poland Digital Product Passport research call, where structured perception is relevant to the Ecodesign for Sustainable Products Regulation [21].

Supporting work should continue in parallel: a formally annotated EgoDex subset would enable precision and recall evaluation of the grounding and mask stages; adaptive windowing should replace fixed stride extraction end to end; and cross dataset evaluation would test generalisability beyond the current source material.

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